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AI/ML intern

## **DAY 4**

### **Task 4**

#### **Testing Body Segmentation Methods**

The objective of this task was to test AI-based body segmentation methods that can separate the human body from the background in real time. This process is important for applications such as: Virtual Try-On systems, Background removal, AR filters, Fitness tracking, Human body analysis , AI-based fashion systems

The testing focused on understanding how segmentation models detect and isolate the human body from surrounding objects and environments.

#### **What is Body Segmentation?**

Body segmentation is a computer vision technique where AI identifies which parts of an image belong to the human body and which parts belong to the background.

The segmentation model performs: Human body detection, Pixel-level classification, Background separation

#### **Simple Representation**

Human Body → Foreground  
Room/Objects → Background

#### **Working Process of Body Segmentation**

The body segmentation system follows several processing stages:

##### **Step 1 — Capture Input Image**

The system captures: Webcam frame, Human body image, Live camera input

This image acts as the input for segmentation processing.

##### **Step 2 — Process Image Using AI Model**

The segmentation model analyzes: Human body shape, Edges, Body contours, Pixel regions

AI then classifies each pixel as:

- Human body
- Background

##### **Step 3 — Generate Segmentation Mask**

The AI creates a mask image where:

- White region → Human body
- Black region → Background

This mask helps isolate the person from the surrounding environment.

#### **Step 4 — Remove Background**

Using the segmentation mask:

- Background pixels are removed
- Only the human body remains visible

This produces the final segmented output.

#### **AI Testing Screenshots**

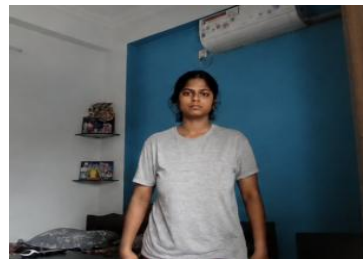
The segmentation testing included four major outputs:

##### **Original Image**

The original webcam frame captured before segmentation.

##### **Purpose**

- Acts as the input image
- Contains full background and human body



##### **Segmentation Mask**



The binary segmentation mask generated by the AI model.

##### **Observation**

- White region represents detected human body
- Black region represents removed background

##### **Importance**

This confirms that the AI correctly identified body regions.

##### **Background Removed Output**

The final segmented image after removing the background.

##### **Observation**

- Human body remains visible
- Background is removed successfully

##### **Importance**

This demonstrates successful body isolation.



Segmentation Testing Analysis

| Observation            | Result     |
|------------------------|------------|
| Human body detection   | Successful |
| Background separation  | Accurate   |
| Real-time segmentation | Smooth     |
| Edge detection         | Good       |
| Body isolation         | Effective  |
| Noise in output        | Minimal    |
| Processing speed       | Fast       |
| Segmentation stability | Stable     |

Observations During Testing

During testing:

- The AI model successfully identified the human body.
- Background removal worked effectively in real time.
- Segmentation remained stable during body movement.
- Clear body contours were detected.
- Minor edge imperfections appeared near hair and arm boundaries.

Advantages of Body Segmentation

| Advantage            | Description                             |
|----------------------|-----------------------------------------|
| Background Removal   | Removes unwanted surroundings           |
| Human Isolation      | Detects only the human body             |
| Real-Time Processing | Fast live segmentation                  |
| AI Fashion Support   | Useful for Virtual Try-On systems       |
| AR Integration       | Supports augmented reality applications |

**Limitations Observed**

| Limitation           | Description                        |
|----------------------|------------------------------------|
| Lighting Sensitivity | Poor lighting reduces accuracy     |
| Edge Imperfections   | Small errors near hair/body edges  |
| Complex Backgrounds  | Crowded scenes affect segmentation |
| Fast Movement        | Rapid motion may reduce stability  |

**Future Improvements**

Future enhancements may include: Deep learning-based segmentation models, Improved edge refinement, Multi-person segmentation, Higher-resolution processing, Real-time AI optimization

**Applications of Body Segmentation**

Body segmentation is widely used in: Virtual Try-On systems, Online fashion platforms, AI fitness applications, Video conferencing filters, Motion tracking systems, Human-computer interaction

**Conclusion**

The body segmentation testing successfully demonstrated how AI models can isolate the human body from the background using computer vision techniques. The segmentation process achieved accurate body detection, smooth real-time performance, and effective background removal. The testing provided valuable understanding of how segmentation methods support advanced AI applications such as Virtual Try-On systems, AR filters, and intelligent fashion technologies.